

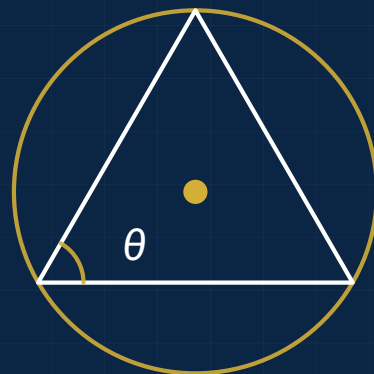


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 7

# Volume & Surface Area

Prism: cross-section  $\times$  length. Cone and pyramid: a third of that. Surface area: add the area of every face you could paint.



4

VARIANTS

22

PAST-PAPER PARTS

65

PRACTICE QUESTIONS

## YOU WILL MASTER

V1 Prisms & cylinders

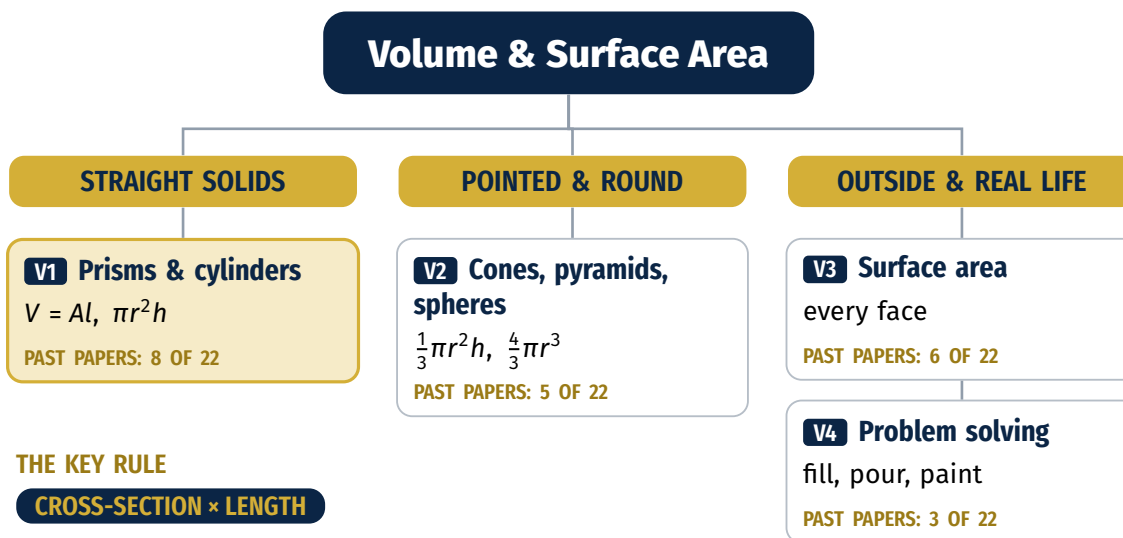
V2 Cones, pyramids & spheres

V3 Surface area

V4 Problem solving

TOPIC MAP • EVERY VARIANT ON ONE PAGE

# Volume at a glance


  
Your notes


## THE KEY RULE

**CROSS-SECTION × LENGTH**

Any prism, any shape of end.

## ★ How often does each variant come up?

20 past-paper questions (Jan 2020 → Jun 2025) hold 22 parts; 65 practice parts — grouped by type.

|                                    | PAST PAPERS (OUT OF 22) | PRACTICE PARTS |
|------------------------------------|-------------------------|----------------|
| <b>V1</b> Prisms, cylinders        | 8                       | 17             |
| <b>V2</b> Cones, pyramids, spheres | 5                       | 15             |
| <b>V3</b> Surface area             | 6                       | 15             |
| <b>V4</b> Problem solving          | 3                       | 18             |

**WHAT THIS TELLS YOU** On every paper: an easy prism/cylinder (3 marks, early) and a harder sphere/cone question (5–6 marks, near the end). The formulas for cones and spheres are on the formula sheet — know when to use them.

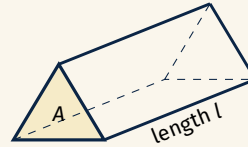
## BEFORE YOU START

## The big idea

**A prism is the same all the way through.**

Slice it anywhere: the same **cross-section**. So volume = **area of the end**  $\times$  **length**.

Pointed solids (cone, pyramid) hold exactly **a third** of the matching prism.



$$V = A \times l$$

 Your notes

### The formulas (cone and sphere are on the formula sheet)

#### Cylinder

$$\pi r^2 h$$

Curved area  $2\pi rh$ .

#### Cone

$$\frac{1}{3}\pi r^2 h$$

Curved area  $\pi rl$ ,  $l$  = slant.

#### Sphere

$$\frac{4}{3}\pi r^3$$

Surface  $4\pi r^2$ .

#### Pyramid

$$\frac{1}{3} \times \text{base} \times h$$

$h$  = vertical height.

### Units

#### Litres

1 litre =  $1000 \text{ cm}^3$   
 $\div 1000$ .

#### $\text{m}^3$ to $\text{cm}^3$

$\times 1\,000\,000$   
 $100^3$ .

#### Hemisphere

$$\frac{2}{3}\pi r^3$$

Surface:  $2\pi r^2$  curved +  $\pi r^2$  flat.



#### CALCULATOR FIRST, THEN CHECK

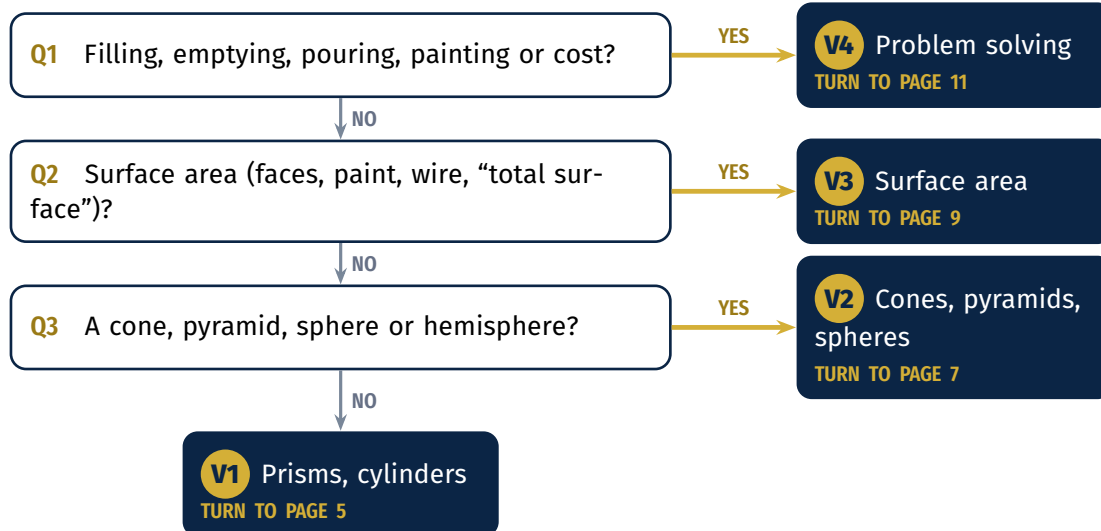
Exact answers: keep  $\pi$  ( $666\pi$ ). Rounded answers: full calculator value, round once.

COUNT / RECOGNISE, THEN CHOOSE

# Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



## COLOUR CODE IN THE WORKING

**Blue** = letter terms  $4x$ ,  $-3y$

**Orange** = plain numbers  $17$ ,  $-13$

**STEP 1** **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

## ON EVERY VARIANT SPREAD

- |                                                                                                     |                                                                                                             |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|  What is it?     |  When you see → you do |
|  How do I do it? |  Watch out             |
|  Worked example  |  Exam tip              |
|  Check yourself  |  Calculator check      |

V1

VARIANT 1 MOST COMMON IN THE EXAM

# Prisms & Cylinders

Area of the cross-section  $\times$  length; cylinder  $\pi r^2 h$ . Backwards: divide.

SEEN IN PAST PAPERS

8 / 22

PRACTICE QUESTIONS: 17

  
 Your notes

**Came up in:** Jan 2020 P2HR Q13 • Jan 2021 P1HR Q5 • May 2021 P2H Q10 • Jun 2023 P2HR Q9 • Nov 2023 P1H Q21 • Nov 2024 P1H Q5 • Jun 2025 P1H Q7

## ? What is it?

### In plain words

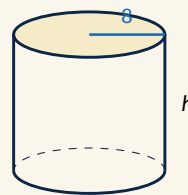
Find the **area of the end face** (triangle, trapezium, circle ...), then multiply by the **length**.

Volume given? **Divide** to get the missing length:  $h = \frac{V}{\pi r^2}$ .

IT LOOKS LIKE

trapezium prism

$$\pi r^2 h = 3892$$



$$\pi(8)^2 h = 3892$$

## How do I do it?

### 1 End face

Name it; find its area.

### 2 Length

The distance between the two ends.

### 3 Multiply

Area  $\times$  length (cylinder:  $\pi r^2 h$ ).

### 4 Backwards

Volume given: divide.

**CROSS-SECTION  $\times$  LENGTH** The most common variant here.

## V1 Prisms & Cylinders

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes

### WHEN YOU SEE → YOU DO

"The volume of the cylinder is 3892 cm<sup>3</sup>. Work out  $h$ "

$$\rightarrow h = 3892 \div (\pi \times 8^2)$$

"Give your answer in litres"

$$\rightarrow \text{cm}^3 \div 1000$$

A triangle on the end with an angle

$$\rightarrow \frac{1}{2}ab \sin C \text{ for the cross-section}$$

### WORKED EXAMPLE PAST PAPER

June 2025 · Paper 1H · Q7 · 3 marks

A solid cylinder has radius 8 cm and height  $h$  cm. Its volume is 3892 cm<sup>3</sup>. Work out  $h$  (1 d.p.).

**STEP 1**  $V = \pi r^2 h$  ◀ cylinder

**STEP 2**  $\pi \times 64 \times h = 3892$  ◀ substitute

**STEP 3**  $h = \frac{3892}{64\pi} = 19.357 \dots$  ◀ divide

**STEP 4**  $\approx 19.4$  ◀ 1 d.p.

**ANSWER**  $h = 19.4$

$$\checkmark \pi \times 64 \times 19.4 \approx 3901 \approx 3892 \checkmark$$

### WATCH OUT

$$\times \pi \times 8 \times h$$

$$\checkmark \pi \times 8^2 \times h$$

Square the radius.

### EXAM TIP

Write the formula with the numbers in before you rearrange.

### Check yourself

calculator first · answers on page 13

**1.1** Dr Eslam

A prism has cross-section 18 cm<sup>2</sup> and length 12 cm. Volume?

Answer: .....

**1.2** Nov 2024 P1H

A cylinder has radius 70 cm and height 18 cm. Volume in litres (nearest litre)?

Answer: .....

**1.3** Jan 2020 P2HR

A prism has a trapezium cross-section (7 and 12 cm, height  $h$ ), length 10 cm, volume 608 cm<sup>3</sup>. Find  $h$ .

Answer: .....

**V2**

VARIANT 2

# Cones, Pyramids & Spheres

Use the formula sheet: cone  $\frac{1}{3}\pi r^2 h$ , sphere  $\frac{4}{3}\pi r^3$ ; hemisphere is half.

SEEN IN PAST PAPERS

5 / 22

PRACTICE QUESTIONS: 15

 Your notes

Came up in: Jan 2020 P2H Q26 • Jun 2022 P1H Q22 • Jun 2023 P2HR Q24 • May 2024 P1H Q23 • Nov 2024 P1H Q22

## ? What is it?

### In plain words

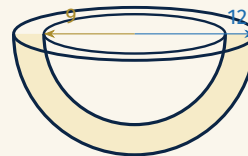
These formulas are **given** on the formula sheet. The skill is choosing the right one and the right **radius**.

A **hemisphere** is half a sphere. A **hollow** bowl = outer hemisphere - inner hemisphere.

### IT LOOKS LIKE

bowl: outer 12

inner 9



$$\frac{2}{3}\pi(12^3 - 9^3)$$

## How do I do it?

**1** Pick

Cone, pyramid, sphere, hemisphere?

**2** Radius

Halve a diameter; check which radius.

**3** FormulaFrom the sheet; hemisphere  $\frac{2}{3}\pi r^3$ .**4** Combine

Add or subtract solids.

**1/3 FOR POINTED SOLIDS**Hemisphere: half of  $\frac{4}{3}$ .

**V2 Cones, Pyramids & Spheres**

VARIANT 2 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"bowl ... outer radius 12 cm, inner radius 9 cm"

→ **Outer – inner hemisphere**

"surface area of A is nine times that of B"

→ **Radius ×3, volume ×27**

"exact value of k" with  $k\pi$

→ **Leave  $\pi$  out of the arithmetic****WORKED EXAMPLE****PAST PAPER**

November 2024 · Paper 1H · Q22 · 3 marks

A hemispherical metal bowl has outer radius 12 cm and inner radius 9 cm. Work out the volume of the metal (nearest whole number).

**STEP 1**  $\frac{2}{3}\pi(12)^3 = 1152\pi$  ◀ outer hemisphere

**STEP 2**  $\frac{2}{3}\pi(9)^3 = 486\pi$  ◀ inner hemisphere

**STEP 3**  $1152\pi - 486\pi = 666\pi$  ◀ subtract

**STEP 4**  $= 2092.3 \dots$  ◀

**ANSWER** 2092 cm<sup>3</sup>

$$\checkmark \frac{2}{3}\pi(1728 - 729) = 666\pi \checkmark$$

**WATCH OUT**

$$\times \frac{2}{3}\pi(12 - 9)^3$$

$$\checkmark \frac{2}{3}\pi(12^3 - 9^3)$$

Cube each radius, then subtract.

**EXAM TIP**

Write everything as a multiple of  $\pi$  until the last line.

**Check yourself**

calculator first · answers on page 13

**2.1**

Dr Eslam

A cone has radius 6 cm and height 10 cm. Volume (3 s.f.)?

Answer: .....

**2.2**

Jun 2023 P2HR

Sphere A has 9 times the surface area of sphere B; the volumes differ by  $117\pi$  cm<sup>3</sup>. Radius of B?

Answer: .....

**2.3**

Jan 2020 P2H

A sector of angle  $x^\circ$  makes a cone of height 25 cm and volume 1600 cm<sup>3</sup>. Find  $x$ .

Answer: .....

**V3**

VARIANT 3

**Surface Area**

List every face, find each area, add. Curved faces:  $2\pi rh$ ,  $\pi rl$ ,  $4\pi r^2$ .

SEEN IN PAST PAPERS

6 / 22

PRACTICE QUESTIONS: 15

  
 Your notes

**Came up in:** Jan 2021 P2H Q17 • Nov 2021 P2H Q22 • Nov 2024 P2H Q6 • Jun 2025 P2HR Q26 • Jun 2025 P2H Q21

## ? What is it?

### In plain words

Surface area is the area of **every face**

— like the paper needed to wrap it.

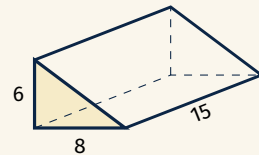
Count the faces: a triangular prism has **2 triangles + 3 rectangles**.

Missing lengths (like the hypotenuse) come from **Pythagoras**.

### IT LOOKS LIKE

triangular prism

hemisphere bowl



$$2 \times 24 + 15(6 + 8 + 10)$$

## How do I do it?

### 1 Faces

List them all (sketch a net).

### 2 Lengths

Pythagoras for slants and hypotenuses.

### 3 Areas

Each face, curved parts by formula.

### 4 Add

Total; “show that”: every stage.

**EVERY FACE, ONCE** Hemisphere: curved + flat circle.

**V3 Surface Area**

VARIANT 3 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"Work out the total surface area of the triangular prism" → **2 triangles + 3 rectangles**

"Show that  $A = 12x^2 + \frac{1690}{x}$ " → **y from the volume, then 6 faces**

"total surface area of the bowl ... in terms of  $\pi$ " → **Outer + inner curved + the rim ring**

**WORKED EXAMPLE** **PAST PAPER**

November 2024 · Paper 2H · Q6 · 3 marks

A solid triangular prism has a right-angled triangle cross-section with sides 6 cm and 8 cm, and length 15 cm. Work out the total surface area.

**STEP 1**  $\sqrt{6^2 + 8^2} = 10$  ◀ third side

**STEP 2**  $2 \times \frac{1}{2} \times 6 \times 8 = 48$  ◀ two triangles

**STEP 3**  $15 \times (6 + 8 + 10) = 360$  ◀ three rectangles

**STEP 4**  $48 + 360 = 408$  ◀ add

**ANSWER** 408 cm<sup>2</sup>

✓ 5 faces counted ✓

**WATCH OUT**

- ✗ one triangle
- ✓ two triangles (both ends)

A prism has two end faces.

**EXAM TIP**

The rectangles together are length × perimeter of the end:  $15 \times 24$ .

**Check yourself**

calculator first · answers on page 13

**3.1**

Dr Eslam

Total surface area of a solid hemisphere of radius 4 cm, in terms of  $\pi$ ?

Answer: .....

**3.2**

Jun 2025 P2H

A cuboid  $3x$  by  $2x$  by  $y$  has volume  $1014 \text{ cm}^3$ . Show the total surface area  $A = 12x^2 + \frac{1690}{x}$ .

Answer: .....

**3.3**

Jun 2025 P2HR

A hemisphere of volume  $6174\pi \text{ cm}^3$  is hollowed to a bowl 2 cm thick. Total surface area of the bowl?

Answer: .....

**V4**

VARIANT 4

## Problem Solving with Volume

Turn the story into volumes (or areas), compare or divide, then answer the question in words.

SEEN IN PAST PAPERS

3 / 22

PRACTICE QUESTIONS: 18

 Your notes

Came up in: Jan 2021 P1H Q3 • Jan 2022 P1HR Q10 • Jun 2025 P1HR Q10

### ? What is it?

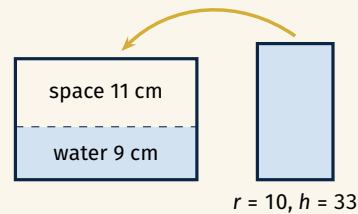
#### In plain words

Water poured from one container into another: the **volume stays the same**. Compare it with the **space** available.

Rates ( $\text{cm}^3$  per minute), tins of paint, costs: one step at a time, with a **conclusion**.

#### IT LOOKS LIKE

pour the cylinder into the cuboid



### How do I do it?

#### 1 Volumes

Each container / the water / the space.

#### 2 Same units

$\text{cm}^3$ , litres,  $\text{m}^3$ .

#### 3 Compare or divide

Fits? How many? How long?

#### 4 Conclude

A sentence with the numbers.

**VOLUME IS KEPT** Compare with the space left.

**V4 Problem Solving with Volume**

VARIANT 4 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

“Show that the cuboid will not be completely full”

→ **Water volume < space left**

“Find the time at which all the water has been pumped out”

→ **Rate from one stage, then total time**

**WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 1HR · Q10 · 3 marks

A cuboid 35 cm by 28 cm by 20 cm holds water 9 cm deep. A full cylinder (radius 10 cm, height 33 cm) is poured in. Show that the cuboid will not be full.

**STEP 1**  $\pi \times 10^2 \times 33 = 3300\pi = 10\,367$

◀ water in the cylinder

**STEP 2+3**  $35 \times 28 \times (20 - 9) = 10\,780$

◀ space above the water

**STEP 4**  $10\,367 < 10\,780$

◀ compare

**ANSWER** it fits — the cuboid is not completely full ✓ 413 cm<sup>3</sup> left over ✓

**WATCH OUT**

✗ compare with the whole cuboid (19 600)

✓ compare with the space left

9 cm of water is already in it.

**EXAM TIP**

“Show that”: both numbers and the comparison sentence are needed.

**Check yourself**

calculator first · answers on page 13

**4.1**

Jan 2022 P1HR

Paint 4 sides and top of a 12 m × 2.5 m × 3 m box; tins cover 15 m<sup>2</sup>; £26.95 a tin after a 10% rise. Cost before?

Answer: .....

**4.2**

Jan 2021 P1H

500 000 cm<sup>3</sup> of water; 75 000 cm<sup>3</sup> pumped out from 10:30 to 12:00, steadily. When is it empty?

Answer: .....

**4.3**

Practice · Medium

12 hemispherical bowls of radius 20 cm are filled from a 200-litre tank. Is there enough water?

Answer: .....

## CHECK YOURSELF

# Answers



Every answer below was checked with sympy (check\_answers.py).

**1.1**  $216 \text{ cm}^3$

**1.2** 277 litres

**1.3**  $h = 6.4$

**2.1**  $377 \text{ cm}^3$

**2.2** 1.5 cm

**2.3**  $x = 107$

**3.1**  $48\pi \text{ cm}^2$

**3.2**  $A = 12x^2 + \frac{1690}{x}$

**3.3**  $1684\pi \text{ cm}^2$

**4.1**  $\text{£}196 < \text{£}200$

**4.2** 20:30

**4.3** No: 201 litres needed

## THE WHOLE TOPIC ON ONE PAGE

# Volume cheat sheet


  
Your notes
**V1 Prisms 8/22**

$$A \times l, \pi r^2 h$$

Backwards: divide.

**V2 Pointed, round 5/22**

$$\frac{1}{3}\pi r^2 h, \frac{4}{3}\pi r^3$$

Formula sheet; hemisphere half.

**V3 Surface 6/22**

$$2 \times 24 + 15 \times 24$$

Every face once.

**V4 Problems 3/22**

$$10\,367 < 10\,780$$

Volume kept; conclude.

**TOP 3 LOST MARKS**

1.  $r$  not squared or cubed ×
2. one end face missing ×
3. no conclusion in “show that” ×

**BOTTOM LINE** Cross-section × length · a third for points · every face once · 1 litre = 1000 cm<sup>3</sup>.